# Srikrishnaa Vadivel

Computational Physicist | First Principles | HPC | Scientific Software

## Summary

Computational physics PhD researcher focused on first-principles calculations of self-trapped excitons, scientific software, and high-performance simulation. Experience includes leadership-class HPC, PETSc/SLEPc, MPI/OpenMP, Python/C++, and ML-assisted materials workflows.

## Research

**Yale University, PhD Researcher** 2021–present

- Perform first-principles calculations of self-trapped excitons using HPC resources at Oak Ridge, Aurora, LBNL, and LLNL.
- Build reproducible Linux, Bash, Python, C/C++, CMake, PETSc/SLEPc, MPI/OpenMP, Docker, and Kubernetes workflows for large simulations.
- Explore machine-learning surrogates and equivariant molecular models for materials and molecular systems.

## Selected Technical Projects

- **DOLFINx PDE Gallery**: weak forms for Poisson, heat, and elasticity, plus solver/visualization scaffold.
- **First-Principles LiF**: plane-wave pseudopotential Hamiltonian and band plotting.
- **Physics Action Notes**: path integrals, imaginary time, EM, GR, QED/QCD, Standard Model, and strings from action principles.
- **Meep Ring Resonator**: photonics FDTD geometry and source setup.

## Industry

**Probe Technologies / Weatherford** 2019–2021

Built CUDA/OpenGL waveform processing and visualization software for acoustic logging tools; also delivered embedded sensor logging firmware and FPGA logic.

## Education

Yale University, PhD Electrical Engineering, 2021–present
Cornell University, MEng ECE, 2017–2018
Cornell University, BA ECE, 2013–2017

## Skills

DFT workflows, PDEs, numerical linear algebra, PETSc, SLEPc, MPI, OpenMP, Python, C++, CMake, Linux, CUDA, PyTorch, DOLFINx concepts, Meep concepts.